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Measuring Factor Crowding When Leverage Is Observable: Evidence from China's A-Share Market

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Abstract

Factor crowding is difficult to measure because holdings are rarely observed and return-based proxies may capture several forms of common movement. We compare residual-return comovement with an observable margin-financing-balance proxy for six style factors in China's A-share market from 2015–2025. Within-factor comovement between the measures is limited. The margin-balance proxy is conditionally negatively associated with subsequent factor returns after controlling for factor momentum, balance levels, long- and short-leg coverage, and factor fixed effects. In the main six-month specification, a one-standard-deviation increase is associated with a 5.2-percentage-point lower horizon-sum return. The association is concentrated in the low-side leg, although subsample and factor-specific estimates are not uniformly significant. Comomentum gives more heterogeneous evidence: a closer weekly Lou–Polk analogue is negatively associated with 12-month compounded momentum returns, but pooled significance is construction-dependent and the association does not survive the strictest multiple-testing corrections. Long-horizon skewness results also vary between pooled factors and momentum alone. Because rolling formation and holding windows overlap, we use pooled circular-shift permutation inference alongside HAC standard errors. The results support a measurement distinction rather than a universal crowding effect: return comovement and leveraged financing are not interchangeable proxies, and all evidence is associational rather than causal.

Keywords: factor crowding, factor decay, comomentum, margin financing, China, A-share market

1. Introduction

Capital flowing into a factor strategy can eliminate the mispricing it exploits and leave crowded trades exposed to reversal (Shleifer and Vishny, 1997; De Long et al., 1990; Stein, 2009; Barberis and Shleifer, 2003). The empirical obstacle is measurement: positions are usually unobserved, while returns also contain the price pressure that crowding creates. The *comomentum* measure

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of Lou and Polk (2022) infers common positioning from residual-return correlations and predicts lower US momentum returns, higher volatility, and more negative skewness.

We compare this indirect proxy with observable margin-financing balances in China's A-share market, where margin-financing arrangements and short-sale eligibility differ from the US setting. The contribution is a measurement comparison, not a single-country replication or a claim to observe total positions. Using six factors over 2015–2025, we find limited comovement between the proxies. Margin balances are negatively associated with pooled future returns; comomentum results vary by horizon, construction, and factor. Neither is uniformly predictive.

Three design concerns are central. Balance changes mechanically contain balance-level information, so we control for that level. Factors with strong past returns may attract financing, so we include and orthogonalise against factor momentum and estimate factor-specific slopes. Finally, overlapping formation and holding windows motivate pooled circular-shift permutation inference with the same estimator used for the observed statistic. These choices connect the paper to work on factor decay and crowding (McLean and Pontiff, 2016; Beck et al., 2016; Lou and Polk, 2022) and Chinese margin trading (Bian et al., 2026; Xing et al., 2026).

2. Data and methodology

2.1. Data

We use the full A-share cross-section from January 2015 to December 2025. Daily prices, trading volumes, turnover, and point-in-time valuation ratios are from Baostock (www.baostock.com); the universe includes delisted firms. The cleaned universe contains 5,418 stocks over 2,674 trading days. After retaining historical ST trading days and filtering ST status only at formation, the monthly panel covers 5,248 stocks. Individual-stock margin-financing balances, covering the designated margin-eligible list, are from the Shanghai and Shenzhen exchanges.

Signals use only information available by month-end, with next-trading-day balance availability. A stratified 40-stock audit supports disclosure-date updating of the valuation fields, without ruling out later revisions. At formation, we exclude stocks that are suspended, ST, financial, or listed for less than 180 days, as well as stock-months with more than ten non-trading days. Industry classification is a current snapshot, a limitation. Portfolios use the full cross-section; expanding margin eligibility affects only the financing proxy, for which we retain coverage controls.

Monthly factors are value (book-to-market), size (negative log float capitalisation), momentum (months $t-12$ to $t-1$), quality (price-to-book divided by trailing price-to-earnings), low volatility (negative prior 60-day volatility), and turnover (negative prior 20-day turnover). Decile membership and float-capitalisation weights are frozen at formation; stocks not trading on the market month-end use their last observed in-month snapshot. At realisation, weights are renormalised over non-missing next-month returns without assigning missing returns a zero. Historical and holding-period ST trading days remain in return construction. A separate raw-daily audit independently verifies this convention and distinguishes observable realisations, in-holding-month terminal events, nonterminal gaps, and sample-end right censoring. Returns are top minus bottom deciles; common-month-end, daily-leg, and terminal-return sensitivities isolate the resulting timing and estimator choices.

2.2. Two measures of crowding

Indirect: comomentum.. For each factor's long and short deciles, the baseline averages sampled pairwise correlations of daily residuals from a local three-factor model over 250 trading days,

capping each leg at 100 stocks with a fixed seed. It is a daily variant, not an exact Lou–Polk replication. A sampling audit repeats the full measure and specification-D inference for 20 predeclared seeds, including the production seed, without pooling seed replicates or selecting by significance. Auxiliary weekly analogues use 52 Friday-to-Friday industry-adjusted returns and correlations with the leave-one-out equal-weighted leg portfolio. Variants require complete histories and impose local price/size screens. Current-snapshot industries and the local benchmark preclude a literal US replication.

Margin-financing-balance proxy.. We aggregate $\Delta MB_{i,t}/MV_{i,t}$ within each leg using formation float-capitalisation weights and subtract the short-leg aggregate from the long-leg aggregate. Snapshots older than 45 calendar days and unavailable lagged snapshots are missing, never zero changes. Intensities are renormalised over valid observations; coverage uses the full formation leg. Size's long-leg coverage is especially thin (mean 0.26 versus 0.82–0.93 elsewhere), motivating two-leg coverage controls and coverage-floor checks. This observable financing proxy is neither an account-level position nor a pure flow.

2.3. Empirical design

For each measure $C_{f,t}$ and horizon $h \in \{1, 3, 6, 12\}$ we estimate

$$R^{f}_{t\to t+h} = \alpha_f + \beta C_{f,t} + \gamma' X_{f,t} + \varepsilon_{f,t}, \tag{1}$$

where $R^{f}_{t\to t+h}$ sums the next h monthly long-short returns from $t+1$, α_f is a factor fixed effect, and X contains past 1-, 6-, and 12-month factor returns, margin level, and two-leg coverage. Auxiliary tests use compounded returns. Newey–West lags are six in primary and twelve in long-horizon specifications; six factor clusters cannot support reliable cluster asymptotics. We also orthogonalise the proxy against past returns and estimate separate factor slopes.

Circular shifts within factor preserve the controls, sample, estimator, and HAC lag length. Reported finite-permutation p-values use $(1 + B_{\geq})/(1 + B)$. Synthetic-null and block-bootstrap power checks accompany the replication. Auxiliary construction and factor families include Benjamini–Hochberg and Bonferroni corrections; individual slopes are exploratory. Primary tests use both baseline proxies at $h \in \{1, 3, 6\}$; horizon 12, alternative constructions, and risk outcomes remain auxiliary.

3. Results

3.1. The two measures of crowding are only weakly related

The two measures have limited linear comovement: within-factor correlations range from -0.28 for value (and -0.15 for turnover) to $+0.25$ for size, and are near zero for quality and low volatility. Pooling the six factors gives -0.13, or -0.07 after removing factor-specific means. Across the 20 seeds, the factor-demeaned pooled Pearson correlation ranges from -0.095 to -0.035; the quality-factor Spearman correlations and the low-volatility-factor correlations can cross zero. These correlations may also reflect timing, coverage, nonlinear dependence, or sampling noise. Figure 1 plots both series for each factor.

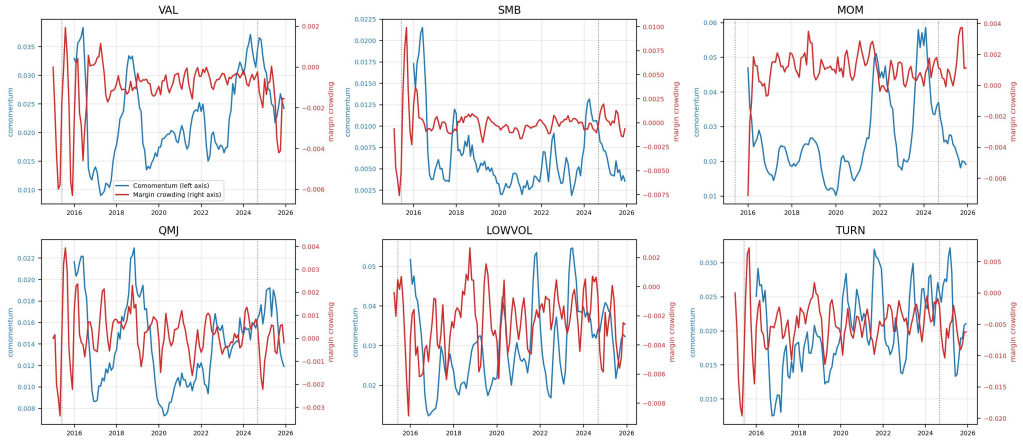


Figure 1: Crowding measures by factor (three-month rolling means). Baseline comomentum (blue, left axis) and margin crowding (red, right axis) use different scales. Dotted lines mark June 2015 and September 2024.

3.2. Return predictability differs by proxy, horizon, and factor

At one, three, and six months, baseline daily comomentum has no robust return association (Table 1). Across all 20 sampling seeds these coefficients remain positive; six-month permutation p-values range from 0.265 to 0.667, with no seed below 0.10. Margin balances are negatively associated with returns. Adding factor momentum, balance level, and two-leg coverage strengthens rather than removes the association: the six-month coefficient changes from −0.0447 with momentum controls to −0.0524 in specification D (permutation $p = 0.010$).

Table 1: Crowding and future factor returns

Measure	h	A	B	C	D
Comomentum	1	0.0011 (0.782)	0.0010 (0.766)	0.0011 (0.720)	0.0038 (0.280)
Comomentum	3	-0.0003 (0.991)	0.0023 (0.790)	0.0026 (0.714)	0.0108 (0.352)
Comomentum	6	0.0010 (0.930)	0.0055 (0.706)	0.0059 (0.686)	0.0208 (0.265)
Comomentum	12	-0.0263 (0.491)	-0.0238 (0.375)	-0.0238 (0.406)	-0.0095 (0.677)
Margin-balance proxy	1	-0.0095*** (0.008)	-0.0169** (0.019)	-0.0177** (0.019)	-0.0190** (0.019)
Margin-balance proxy	3	-0.0202** (0.017)	-0.0280*** (0.010)	-0.0302*** (0.010)	-0.0338*** (0.010)
Margin-balance proxy	6	-0.0252** (0.018)	-0.0447*** (0.010)	-0.0479*** (0.010)	-0.0524*** (0.010)
Margin-balance proxy	12	-0.0261* (0.093)	-0.0484** (0.021)	-0.0530** (0.031)	-0.0574** (0.021)

Notes: Each cell reports the coefficient, starred by the finite circular-shift permutation p-value, with the permutation p-value in parentheses. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The magnitude is economically material: in specification D at $h = 6$, a one-standard-deviation increase in the margin-balance proxy accompanies approximately a 5.2 percentage

point lower horizon-sum long-short return over six months (4.4 points per within-factor standard deviation). At $h = 12$, the compounded-return specification gives $\beta = -0.0709$ with HAC(12) and permutation $p = 0.021$. However, the momentum-factor slope is near zero ($p = 0.667$), while quality is the strongest individual contributor ($p = 0.009$; six-factor BH $p = 0.056$). These are conditional, heterogeneous associations, not universal or causal effects.

The weekly comomentum analogues qualify the shorter-horizon null rather than overturning it. For momentum's 12-month compounded return, the weekly leave-one-out measure gives $\beta = -0.1885$ and permutation $p = 0.031$; low volatility gives $p = 0.048$. Neither survives six-factor BH correction (both $p = 0.143$), and the seven-construction Bonferroni-adjusted momentum result is $p = 0.219$. Pooled 12-month estimates are also construction-dependent ($p = 0.073$ for the core weekly analogue and 0.010 for its screened version). We therefore interpret the long-horizon comomentum evidence as exploratory and factor-specific, not as a general replication of Lou–Polk return predictability.

3.3. The signal is in the low-side leg

Table 2 decomposes the six-month association with common specification-D controls. The long leg contributes little ($\beta = 0.0003$, $p = 0.990$), whereas the short leg carries the spread result ($\beta = 0.0527$, $p = 0.010$). Its own financing intensity also predicts its own return ($\beta = -0.0608$, $p = 0.020$); the long-leg analogue is insignificant ($p = 0.196$). Comomentum shows no corresponding leg pattern.

Table 2: Leg decomposition of the crowding association

Panel	Measure	Return leg	h	β	HAC t	Perm. p	N
A	Comomentum	Long-short	6	0.0208	2.11	0.265	669
	Comomentum	Long leg	6	0.0052	0.62	0.696	669
	Comomentum	Short leg	6	-0.0156	-1.28	0.608	669
A	Margin-balance proxy	Long-short	1	-0.0190**	-3.44	0.019	699
	Margin-balance proxy	Long leg	1	-0.0020	-0.64	0.533	699
	Margin-balance proxy	Short leg	1	0.0170**	3.45	0.019	699
A	Margin-balance proxy	Long-short	6	-0.0524***	-4.32	0.010	669
	Margin-balance proxy	Long leg	6	0.0003	0.03	0.990	669
	Margin-balance proxy	Short leg	6	0.0527***	4.40	0.010	669
B	Margin-balance proxy, long leg	Long leg	1	-0.0093*	-1.91	0.093	699
B	Margin-balance proxy, short leg	Short leg	1	-0.0253***	-4.03	0.009	699
B	Margin-balance proxy, long leg	Long leg	6	-0.0184	-1.61	0.196	669
B	Margin-balance proxy, short leg	Short leg	6	-0.0608**	-3.96	0.020	669

Notes: Panel A regresses each leg's horizon-sum return on the total margin-balance or comomentum measure, with the identical controls and sample as specification D, so the long-short coefficient equals the long-leg coefficient minus the short-leg coefficient. Panel B pairs each leg's return with that leg's own financing measure. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.4. Flows versus stocks

Flow variants scale monthly financing purchases, or purchases less repayments where available, by float capitalisation. Purchase flows are significant only at one month; Shanghai net flows are significant at all four horizons (Table 3). In horse races, purchase flows add no independent content, while net flows retain content only at six months. Balance changes correlate 0.546 and 0.540 with the two flows, indicating partly distinct signals.

Table 3: Financing flows versus balance changes

Measure	h	β	HAC t	Perm. p	N
Financing purchases (buy flow)	1	-0.0378**	-2.91	0.031	644
Financing purchases (buy flow)	3	-0.0571*	-2.81	0.097	632
Financing purchases (buy flow)	6	-0.0811	-2.79	0.169	614
Financing purchases (buy flow)	12	-0.0572	-1.44	0.472	578
Net financing flows (SSE)	1	-0.0178**	-3.43	0.016	644
Net financing flows (SSE)	3	-0.0302**	-3.19	0.048	632
Net financing flows (SSE)	6	-0.0623**	-5.23	0.017	614
Net financing flows (SSE)	12	-0.0529**	-2.33	0.019	578

Notes: Both measures are leg-level float-weighted monthly financing intensities (long leg minus short leg) with specification-D controls and the measure's own trailing twelve-month mean as an additional level control; net flows are available for Shanghai only. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5. Risk diagnostics are mixed and do not validate either proxy

Using common samples and HAC(12), pooled weekly-skewness coefficients for the core weekly analogue are -0.0965 at six months and -0.0870 at twelve months (permutation $p = 0.010$; four-measure BH $p = 0.020$ and 0.021). Yet twelve-month daily volatility and bad-week frequency give $p = 0.667$ and 0.188. For momentum alone, six- and twelve-month weekly-skewness p-values are 0.775 and 0.729; only its three-month daily-volatility result is positive ($p = 0.010$, BH $p = 0.038$). These mixed diagnostics neither establish a failed implementation nor validate comomentum as crowded arbitrage trading. Financing likewise lacks a uniform second-moment pattern.

3.6. Separating crowding from factor momentum

Orthogonalising against past returns and margin level leaves the primary margin-balance association intact (Table 4). Six factor-specific slopes remain exploratory.

3.7. Tail evidence depends on the outcome and horizon

One-month margin-balance quantile coefficients are largest near the median and upper quantiles; the 10% left tail is insignificant ($p = 0.878$). Baseline comomentum is insignificant at all reported quantiles. These are not formal cross-quantile tests. At twelve months, however, margin balances predict a higher pooled share of weeks below -5% ($\beta = 0.0095$, $p = 0.010$; BH $p = 0.042$), but not for momentum alone ($p = 0.938$). Weekly comomentum's pooled skewness lacks matching bad-week evidence. No single tail statistic therefore supports a general interpretation.

Table 4: Separating crowding from factor momentum

Measure	Layer	h	β	HAC t	Perm. p
Comomentum	Layer 1: orthogonalised residual	1	0.0011	0.30	0.729
Comomentum	Layer 1: orthogonalised residual	3	0.0027	0.35	0.781
Comomentum	Layer 1: orthogonalised residual	6	0.0062	0.53	0.735
Comomentum	Layer 3: all controls simultaneously	1	0.0011	0.30	0.720
Comomentum	Layer 3: all controls simultaneously	3	0.0026	0.36	0.714
Comomentum	Layer 3: all controls simultaneously	6	0.0059	0.52	0.686
Margin-balance proxy	Layer 1: orthogonalised residual	1	-0.0177***	-3.22	0.009
Margin-balance proxy	Layer 1: orthogonalised residual	3	-0.0299***	-3.43	0.010
Margin-balance proxy	Layer 1: orthogonalised residual	6	-0.0473***	-3.85	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	1	-0.0177**	-3.27	0.019
Margin-balance proxy	Layer 3: all controls simultaneously	3	-0.0302***	-3.75	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	6	-0.0479***	-4.03	0.010
Comomentum	Layer 2: cross-factor mean slope	1	-0.0038	-0.80	—
Margin-balance proxy	Layer 2: cross-factor mean slope	1	-0.0158	-2.23	—

Notes: Layer 2 reports cross-factor mean slopes; Margin-balance proxy factor-specific slopes are negative for 5/6 factors (2 significant at the 10% level); Comomentum factor-specific slopes are negative for 4/6 factors (0 significant at the 10% level); factor-level rows are in the replication outputs. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.8. Robustness and the scope of individual null results

Margin-balance coefficients are negative in all three subsamples but never individually significant at 10% ($p = 0.139$–0.167; Table 5). Results survive equal weighting, compounded returns, excluding thinly covered size, and coverage floors. Six- and 24-month balance-change coefficients remain negative but weaken (permutation $p = 0.042$ and 0.274). Changing the stale-snapshot cutoff from 45 to 90 calendar days leaves all four horizon estimates virtually unchanged. Coefficients are negative above and below factor-specific comomentum medians; their interaction is marginal ($p = 0.088$), precluding an independence claim. Block-bootstrap minimum detectable effects of 13.1–14.0% annualised apply only to baseline short-horizon comomentum null cells, not a global comomentum null.

Replacing monthly valid-return renormalisation with compounded frozen-weight daily legs gives a six-month margin-balance coefficient of -0.0521 ($p = 0.010$), versus -0.0524 in production. A partial common-month-end variant gives -0.0525 ($p = 0.010$). Because omitted terminal payoffs can bias returns (Shumway, 1997), the pre-switch audit recovered 722 missing stock-month returns and materially changed 446 observed returns by more than one basis point, all with subsequent ST trading days. After adopting the audited convention in production, independent raw-daily reconstruction leaves no remaining differences above one basis point. The current sample identifies 172 in-holding-month terminal events; 63 lack an observable monthly return, of which one is sample-right-censored and is not imputed. Raw realisations give a six-month coefficient of -0.0524; zero and -100% terminal-payoff stresses give -0.0524 and -0.0522, respectively (all $p = 0.010$). These stresses are scenario bounds, not estimated Chinese delisting or share-exchange payoffs. Table 6 reports both proxies and all horizons for the formation variants, and the core six-month terminal-return specifications; the accompanying CSV contains every terminal-return horizon.

Table 5: Robustness

Check	Grouping	Measure	β	HAC t	Perm. p	N
Subsample	2015–2018 leverage cycle	Comomentum	-0.0036	-0.63	0.361	216
Subsample	2015–2018 leverage cycle	Margin-balance proxy	-0.0050	-1.40	0.167	272
Subsample	2019–2021 foreign inflows	Comomentum	0.0033	0.49	0.722	216
Subsample	2019–2021 foreign inflows	Margin-balance proxy	-0.0246	-2.14	0.139	216
Subsample	2022–2025 retail margin	Comomentum	0.0093	1.52	0.149	282
Subsample	2022–2025 retail margin	Margin-balance proxy	-0.0142	-2.14	0.149	282
Equal-weighted	Equal-weighted	Comomentum	-0.0002	-0.08	0.950	714
Equal-weighted	Equal-weighted	Margin-balance proxy	-0.0074***	-2.53	0.008	770
Window	Balance change, 6 months	Margin-balance proxy	-0.0133**	-2.06	0.042	765
Window	Balance change, 24 months	Margin-balance proxy	-0.0062	-1.29	0.274	757
Variant	Long leg only	Margin-balance proxy, long leg	-0.0008	-0.31	0.807	771
Variant	Short leg only	Margin-balance proxy, short leg	0.0029	1.35	0.336	770

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Formation and return-realisation sensitivity

Variant	Measure	h	β	HAC t	Perm. p	N
Frozen-weight daily legs	Comomentum	1	0.0034	0.98	0.364	699
	Comomentum	3	0.0102	1.56	0.371	687
	Comomentum	6	0.0196	2.01	0.284	669
	Comomentum	12	-0.0107	-0.58	0.646	633
Frozen-weight daily legs	Margin-balance proxy	1	-0.0188**	-3.47	0.019	699
	Margin-balance proxy	3	-0.0333***	-4.17	0.010	687
	Margin-balance proxy	6	-0.0521***	-4.35	0.010	669
	Margin-balance proxy	12	-0.0575**	-2.60	0.021	633
Common month-end (partial)	Comomentum	1	0.0037	1.02	0.308	699
	Comomentum	3	0.0105	1.57	0.362	687
	Comomentum	6	0.0204	2.08	0.265	669
	Comomentum	12	-0.0098	-0.53	0.667	633
Common month-end (partial)	Margin-balance proxy	1	-0.0189**	-3.42	0.019	699
	Margin-balance proxy	3	-0.0339***	-4.22	0.010	687
	Margin-balance proxy	6	-0.0525***	-4.39	0.010	669
	Margin-balance proxy	12	-0.0578**	-2.61	0.021	633
Raw realisations incl. ST days	Comomentum	6	0.0208	2.11	0.265	669
Raw realisations incl. ST days	Margin-balance proxy	6	-0.0524***	-4.32	0.010	669
Terminal 0% stress	Comomentum	6	0.0208	2.10	0.265	669
Terminal 0% stress	Margin-balance proxy	6	-0.0524***	-4.32	0.010	669
Terminal -100% stress	Comomentum	6	0.0208	2.11	0.265	669
Terminal -100% stress	Margin-balance proxy	6	-0.0522***	-4.32	0.010	669

Notes: Specification-D estimates; production estimates are in the main return table. Past-return controls and forward outcomes are recomputed from each return series. The daily-leg variant compounds available frozen-weight contributions. The partial common-month-end variant changes membership and returns but retains production crowding and coverage controls. Raw realisations retain subsequent ST trading days. Terminal scenarios apply only to unobserved in-holding-month terminal events and are not estimated payoffs; all horizons are in the accompanying regression CSV. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4. Conclusion

The margin-balance proxy is negatively associated with pooled future returns, with the association concentrated in the low-side leg, but it is not uniformly significant across factors or subsamples. Baseline daily comomentum has little short-horizon association; weekly analogues produce exploratory long-horizon results that are construction-dependent and weakened by multiple-testing corrections. Pooled skewness and bad-week findings likewise do not generalise to momentum alone. The defensible conclusion is a measurement distinction, not a universal crowding effect or validation of either proxy. All estimates are associational. Longer samples, historical industries, position-level data, and quasi-experimental eligibility variation are needed for stronger identification.

Author declarations

Data availability

The daily market and valuation data are obtained from Baostock, and the margin-financing data are obtained from public exchange disclosures. The analysis code, configuration, and manuscript materials are available at `https://github.com/WEN365-1/a-share-factor-crowding-research`. No stand-alone raw, security-level, aggregate, or other provider-derived datasets are redistributed while asset-specific redistribution permission remains unresolved. Researchers must obtain the upstream inputs under the providers' terms and construct the documented analysis inputs themselves; the public package alone does not reproduce the empirical results.

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CRediT authorship contribution statement

Jiaheng Wen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used DeepSeek and Kimi to improve readability and language. The author also used these tools for research assistance and code and methodological review, independently verified the reported results, reviewed and edited all content, and takes full responsibility for the content of the published article.

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